

Disproof of TLMC Conjecture 00000000462: Spanning Trees of the Circulant Graph $C_n(1, 2)$

TLMC submission

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Abstract

TLMC Conjecture 00000000462 claims that $\tau(C_n(1, 2))$, the number of spanning trees of the circulant graph on n vertices with jumps 1 and 2, satisfies a linear recurrence whose largest real characteristic root tends to α^2 with $\alpha = 2 + \sqrt{3}$, i.e. $(2 + \sqrt{3})^2 = 7 + 4\sqrt{3} \approx 13.93$. We evaluate the sequence exactly,

$$\tau(C_n(1, 2)) = \frac{n(L_{2n} - 2(-1)^n)}{5} \quad (n \geq 5),$$

with L_m the Lucas numbers, certified against the matrix–tree theorem by exact integer determinants. Consequently $\tau^{1/n} \rightarrow \varphi^2 = (3 + \sqrt{5})/2 \approx 2.6180$, and every Laplacian eigenvalue is at most $25/4 = 6.25 < 8 < 7 + 4\sqrt{3}$. The claimed limit $\alpha^2 \approx 13.93$ is impossible; the conjecture is **false**.

1 The conjecture

Conjecture 1 (00000000462). $\tau(C_n(1, 2))$ satisfies a linear recurrence whose largest real characteristic root tends to α^2 , where $\alpha = 2 + \sqrt{3}$ (“directly verifiable against the matrix–tree theorem, then stateable in closed form”).

Throughout, $C_n(1, 2)$ is the circulant graph on vertices $\mathbb{Z}_n$ in which i is adjacent to $i \pm 1$ and $i \pm 2$; this is a simple graph exactly for $n \geq 5$, the regime relevant to the conjecture.

2 Exact evaluation

For $n \geq 5$ the Laplacian eigenvalues of $C_n(1, 2)$ are

$$\lambda_k = 2\left(1 - \cos \frac{2\pi k}{n}\right) + 2\left(1 - \cos \frac{4\pi k}{n}\right) = 2(1 - c_k)(3 + 2c_k), \quad c_k = \cos \frac{2\pi k}{n}, \quad k = 1, \dots, n-1.$$

Using the product formula $\prod_{k=0}^{n-1} (x - c_k) = 2^{1-n}(T_n(x) - 1)$ for the Chebyshev polynomial T_n , and $T_n(-\frac{3}{2}) = \frac{(-1)^n}{2}L_{2n}$ (because $(-\frac{3}{2} \pm \frac{\sqrt{5}}{2})^n = (-\varphi^{\mp 2})^n$ and $\varphi^{2n} + \varphi^{-2n} = L_{2n}$), the matrix–tree theorem yields

Proposition 1. For all $n \geq 5$,

$$\tau(C_n(1, 2)) = \frac{1}{n} \prod_{k=1}^{n-1} \lambda_k = \frac{1}{n} \cdot n^2 \cdot \frac{L_{2n} - 2(-1)^n}{5} = \frac{n(L_{2n} - 2(-1)^n)}{5}.$$

Remark 1 (Certification). Proposition 1 was certified computationally, independently of the Chebyshev algebra: exact integer Bareiss determinants of the Laplacian minor reproduce the closed form for $n = 5, 6, 7, 8, 9, 10, 12, 15, 20, 24$, and eigenvalue products match it for $n = 40, 80$ with relative error below 10^{-14} (script `reproduce.py` in the submission).

n	$\tau(C_n(1, 2))$	$\tau^{1/n}$	
5	125	2.626528	
8	3,528	2.776138	
15	5,581,500	2.816978	(peak region)
20	915,304,500	2.805939	(verifier: 2.81)
40	418,891,171,182,561,000	2.757735	(verifier: 2.76)
80	4.3867×10^{34}	2.710359	(verifier: 2.71)

Table 1: Exact values from Proposition 1; the growth rate is already decreasing toward $\varphi^2 \approx 2.6180$.

3 The true growth rate

Since $L_m = \varphi^m + \psi^m$ with $\psi = -\varphi^{-1}$, $|\psi| < 1$, we have $\varphi^{2n} \leq L_{2n} \leq \varphi^{2n} + 1$, so Proposition 1 gives the squeeze

$$\varphi^2 \left(\frac{n}{5}\right)^{1/n} \leq \tau(C_n(1, 2))^{1/n} \leq \varphi^2 \left(1 + \varphi^{-2n}\right)^{1/n} \left(\frac{n}{5}\right)^{1/n},$$

hence

$$\lim_{n \rightarrow \infty} \tau(C_n(1, 2))^{1/n} = \varphi^2 = \frac{3 + \sqrt{5}}{2} \approx 2.618034. \quad (1)$$

Moreover $n \mapsto \tau(C_n(1, 2))/n$ is a C-finite sequence with the Lucas recurrence, whose only characteristic roots are φ^2 and $-\varphi^{-2}$; thus the largest real characteristic root of the natural recurrence is φ^2 for *every* n , not merely in the limit.

4 Why α^2 is impossible

Each factor $2(1 - \cos \theta) \in [0, 4]$, so every Laplacian eigenvalue satisfies $\lambda_k \leq 8$; in fact the sharp bound is

$$\sup_{\theta} (4 - 2 \cos \theta - 2 \cos 2\theta) = \frac{25}{4} = 6.25, \quad \text{attained at } \cos \theta = -\frac{1}{4}.$$

The matrix–tree theorem therefore gives, for every $n \geq 5$,

$$\tau(C_n(1, 2)) \leq \frac{1}{n} \left(\frac{25}{4}\right)^{n-1} \implies \tau(C_n(1, 2))^{1/n} < \frac{25}{4} = 6.25 < 8 < \alpha^2,$$

where $\alpha^2 = 7 + 4\sqrt{3} > 8$ because $\sqrt{3} > \frac{1}{4}$. This contradicts Conjecture 00000000462 under its stated reading: a limit $\tau^{1/n} \rightarrow \alpha^2 \approx 13.93$ “directly verifiable against the matrix–tree theorem” cannot hold when $\tau^{1/n} < 6.25$ for all n and $\tau^{1/n} \rightarrow \varphi^2 \approx 2.618$ by (1). The separator is rigorous at the level of integers:

$$\varphi^2 < 3 < \frac{25}{4} < 8 < 9 < \alpha^2 < 14,$$

using $\sqrt{5} < 3$ (as $5 < 9$) and $\frac{1}{4} < \sqrt{3} < \frac{7}{4}$ (as $1 < 48 < 49 = (4\sqrt{3})^2$'s comparison against 1^2 and 7^2).

All decisive numerical links are machine-checked in core Lean 4 (no Mathlib, no axioms): the exact values of τ for $n \in \{5, \dots, 80\}$, the sandwich $5^{80} < \tau(80) 2^{80} \wedge \tau(80) < 8^{80}$ (growth rate in $(5/2, 8)$ at $n = 80$), the ceiling $a, b \leq 4 \Rightarrow a + b \leq 8$, and the integer skeletons $1 < 48 < 49$, $4 < 5$, $5 < 9$. The audit (`lake env lean Check.lean`) reports “*does not depend on any axioms*” for all 25 declarations; there are no `sorry`s.

5 Boundary cases and remarks

- The closed form of Proposition 1 requires $n \geq 5$: for $n \leq 4$ the jumps $1, 2, -1, -2$ collide mod n . E.g. at $n = 4$ the graph is K_4 with $\tau = 16$, while the formula would give 36; at $n = 3$ one obtains a multigraph. These degeneracies are irrelevant to the asymptotic conjecture.
- The peak of $\tau^{1/n}$ near $n \approx 15$ explains why a naive fit of early data could suggest a rate well above φ^2 ; it remains below 2.82, still far from 13.93.
- The conjecture’s failure is structural: the tree entropy of a one-dimensional circulant is an integral average of $\log \lambda(\theta)$, which cannot exceed $\log \max_{\theta} \lambda(\theta) = \log 6.25 < \log \alpha^2$.

6 Verdict

Conjecture 00000000462 is FALSE. The true exponential growth rate is $\varphi^2 = (3 + \sqrt{5})/2 \approx 2.618$, with exact closed form $\tau(C_n(1, 2)) = n(L_{2n} - 2(-1)^n)/5$ for $n \geq 5$; the claimed root limit $(2 + \sqrt{3})^2 \approx 13.93$ exceeds the universal eigenvalue ceiling $25/4$.